CodeEmy™
TECHNOLOGIES
Ignite the Flame of Education

Clinical SAS Programming

Master the Clinical SAS Programming

Duration: 3 Months | Mode: Online | Level: Beginner to Advanced
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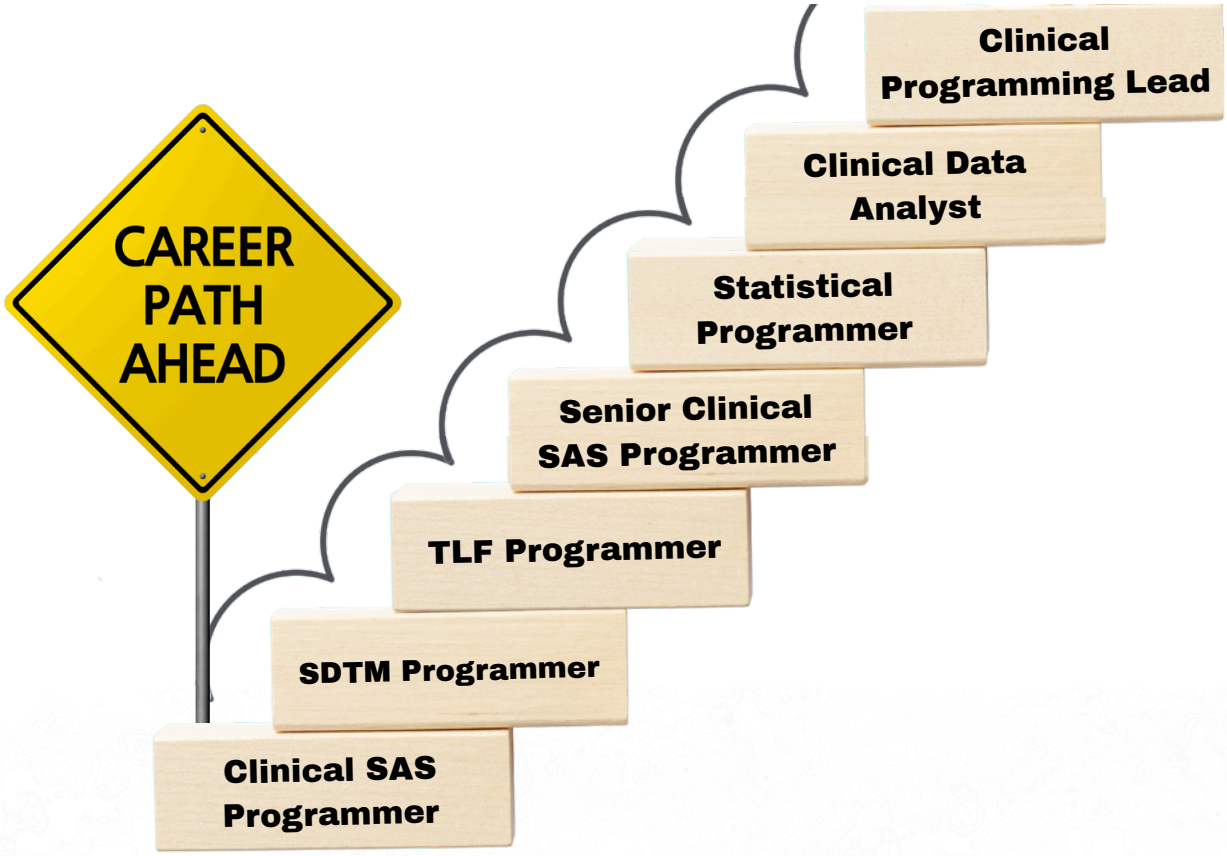
Scope of
Clinical SAS & Clinical Programming Industry

- Clinical Trial Data Analysis
- Pharmaceutical & Biotechnology Industry
- Clinical Research Organizations (CROs)
- Clinical Data Reporting
- Statistical Programming
- Clinical Data Standards
- Regulatory Submission Programming
- Clinical Trial Tables, Listings & Figures
- Clinical Research Data Analysis
- Drug Development & Regulatory Reporting

What Students Will Learn

- Clinical Research → SAS Fundamentals → Data Manipulation → PROC SQL → Macros → Clinical Data → CDISC → SDTM → ADaM → TLF → Statistical Analysis → Regulatory Reporting

Here is all the Career Fields you can excel at



CAREERS

CodeEmy TECHNOLOGIES

Overview:

Curriculum

www.codeemy.in

Week 1

- Introduction to Clinical Research & Drug Development Lifecycle
- Clinical Trial Phases, Sponsor, CRO & Clinical Sites
- Role of Statistical Programming, Introduction to SAS
- SAS Environment, SAS Libraries & Datasets
- Variables & Observations, SAS Data Types

Project: Clinical Trial Data Lifecycle Mapping

Week 2

- SAS Program Structure & DATA Step
- INPUT & CARDS & Variables & Formats
- Character & Numeric Data & Operators
- Conditional Statements & IF / THEN / ELSE
- DO Loops, Functions & Missing Values

Project: Clinical Patient Dataset Creation

Week 3

- Reading External Data & Importing Data
- Data Cleaning, Sorting Data & Filtering Observations
- Creating Variables, Combining Datasets & SET Statement
- MERGE Statement, PROC SORT & PROC CONTENTS

Project: Clinical Dataset Cleaning

Week 4

- PROC PRINT, PROC SORT & PROC FREQ
- PROC MEANS, PROC SUMMARY & PROC TRANSPOSE
- PROC FORMAT & PROC REPORT
- Descriptive Statistics & Frequency Analysis

Project: Patient Demographic Analysis

Week 5

- Introduction to PROC SQL & SELECT Statements
- WHERE Conditions, GROUP BY, ORDER BY
- CASE Statements, Joins, Inner Join & Left Join
- Right Join, Subqueries & SQL-Based Data Validation

Project: Clinical Data Query & Analysis

Week 6

- Introduction to Macros
- Macro Variables, Functions & Parameters
- Conditional Macro Processing & Iterative Macro Processing
- Reusable Programs & Macro Debugging
- Automation Concepts

Project: Automated Clinical Reporting Macro

Week 7

- Clinical Trial Dataset Structure & Clinical Data Lifecycle
- Introduction to CDISC & CDASH Fundamentals
- SDTM Fundamentals & SDTM Domains
- Variables, Controlled Terminology, Standardization Concepts & Raw Data to Standardized Data

Project: Raw Clinical Data to CDISC Mapping

Week 8

- Introduction to SDTM Programming
- SDTM Domain Structure & Subject-Level Data
- Demographics — DM & Adverse Events — AE
- Concomitant Medications — CM & Exposure — EX
- Laboratory — LB & Vital Signs — VS
- Medical History — MH & Data Mapping

Project: Build Sample SDTM Domains

Week 9

- SDTM Mapping Techniques & Domain Relationships
- Variables & Metadata, Controlled Terminology
- SDTM Validation Concepts, Data Quality Checks
- Cross-Domain Validation, Missing Data Checks
- Duplicate Records, Programming Standards

Project: SDTM Data Validation Report

Week 10

- Introduction to ADaM & ADaM Principles
- Analysis Data Structure, ADSL & BDS Concepts
- Analysis Variables, Treatment Variables & Population Flags
- Baseline & Change from Baseline, Derived Variables & Traceability

Project: Create a Sample ADSL Dataset

Week 11

- Introduction to Statistical Analysis & Descriptive Statistics
- Mean, Median & Standard Deviation, Frequency Analysis
- Confidence Intervals, Basic Statistical Tests, T-Test Concepts
- ANOVA Concepts, Regression Concepts, Tables, Listings, Figures
- PROC REPORT, ODS

Project: Clinical Trial TLF Generation

Week 12

- Clinical Programming Workflow & Raw Data → SDTM → ADaM
- ADaM → Statistical Analysis, TLF Generation, Quality Control
- Programming Validation, Regulatory Submission Concepts
- Define-XML Overview, Documentation, Resume Preparation, Clinical SAS Interview Preparation

Project: End-to-End Clinical SAS Capstone

Hands - On Projects

Mini Projects (Beginner to Intermediate Level)

1. Clinical Patient Data Analysis
Skills Used: SAS DATA Step, PROC SORT, PROC FREQ, PROC MEANS
Goal: Import and analyse a simulated clinical patient dataset and generate demographic summaries.

2. Clinical Dataset Cleaning & Transformation
Skills Used: DATA Step, functions, conditional logic, PROC SQL
Goal: Identify missing, inconsistent and duplicate records and create a clean clinical dataset.

3. SDTM Domain Mapping
Skills Used: CDISC, SDTM, data mapping, SAS programming
Goal: Convert simulated raw clinical data into selected SDTM domains such as DM, AE, CM and LB.

Major Projects (Advanced Level)

1. SDTM Programming Project
Skills Used: SAS, CDISC, SDTM, data transformation, validation
Goal: Transform raw clinical trial datasets into standardized SDTM datasets and perform validation checks.

2. ADaM & Statistical Analysis Project
Skills Used: SAS, ADaM, ADSL, analysis variables, statistical procedures
Goal: Create an analysis-ready dataset and perform basic statistical analysis on simulated clinical trial data.

3. End-to-End Clinical SAS Capstone
Skills Used: SAS Programming, PROC SQL, Macro, SDTM, ADaM, TLF, validation
Goal: Complete a simulated clinical programming workflow

Tools

You will learn



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your learning journey

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Certificate of Internship Completion from CODEEMY

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Contact Us

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